

Descriptions of Generalized IS Functions

gen_IS_fun

Generalized function for running plug-in posteriors, IIS, and AIS.

Usage

```
gen_IS_fun(y,
           x = NULL,
           sample_params,
           init_params,
           lik_fun = NULL,
           model = c("AIS", "IIS", "plugin", "AIS_copula"),
           zeta_ppd,
           locs = NULL,
           R = 100,
           m_vecchia = 20,
           mcmc_samples = 1000,
           burn_in = 1000)
```

Arguments

- **y**: A numeric vector of length **n** containing outcome data
- **x**: A **n** × **p** matrix representing known fixed-effect covariate predictors. Defaults to **NULL**
- **sample_params**: A function used to sample model-level parameters. Inputs **y**, **x**, current draw **zeta_sam**, and a named list **params** containing current draw of model-level parameters we are conditioning on.
- **init_params**: A function returning a named list of initial values for model parameters.
- **lik_fun**: A function evaluating an **S** × **n** log-likelihood matrix of **y** conditioned on Stage 1 partial posterior draws and current parameters. Defaults to **NULL** and is not used by the plug-in method.
- **model**: A character string specifying the sampling algorithm. Accepted options are:
 - “plugin”: Fixed spatial latent process, set to Stage 1 posterior means by default.
 - “IIS”: Independent Importance Sampling.
 - “AIS”: Adjusted Importance Sampling.
 - “AIS_copula”: Adjusted Importance Sampling with continuous Empirical CDF transformation, spatial Gaussian Copula, and Vecchia likelihood re-weighting.
- **zeta_ppd**: A numeric matrix of dimension **S** × **n** containing **S** posterior predictive draws of unobserved variable ζ obtained from Stage 1.
- **locs**: A numeric matrix of dimension **n** × 2 specifying spatial coordinates (x_1, x_2) for each location. Used only when **model** = “AIS_copula”.
- **R**: An integer defining the number of discrete candidate draws resampled in the second sampling stage of AIS. Defaults to 300.
- **m_vecchia**: An integer defining the maximum number of nearest neighbors utilized in the Vecchia approximation for spatial covariance likelihood calculation. Defaults to 20. Used only when **model** = “AIS_copula”.
- **mcmc_samples**: An integer specifying the total number of MCMC iterations. Defaults to 1000.

- **burn_in**: An integer specifying the number of initial MCMC iterations to discard as burn-in. Defaults to 1000.

Value

A named list containing post-burn-in MCMC samples (length M):

- **theta**: A numeric vector of length M containing posterior draws for the coefficient θ for the unobserved predictor ζ .
- **beta**: A numeric matrix of dimension M $\times$ p containing posterior draws for known covariate regression parameters X (returned if **x** is provided).
- **sigma_y_sq / r**: A numeric vector of length M containing posterior draws for residual observation variance σ_ϵ^2 (Normal) or dispersion size parameter r (Negative Binomial).
- **zeta_sam**: A numeric matrix of dimension M $\times$ n containing posterior samples of ζ .

Other outputs can exist for custom **sample_params**, **init_params**, and **lik_fun**.

init_params_normal

Generates initial starting values for regression coefficients for known predictors (β), the regression coefficient for unobserved ζ (θ), and the Gaussian residual variance (σ_ϵ^2).

Usage

```
init_params_normal(y, x = NULL, zeta_ppd)
```

Arguments

- **y**: A numeric vector of length n containing outcome data
- **x**: A n $\times$ p matrix representing known fixed-effect covariate predictors. Defaults to NULL
- **zeta_ppd**: A numeric matrix of dimension S $\times$ n containing S posterior predictive draws of unobserved variable ζ obtained from Stage 1.

Value

A named list containing: * **beta**: Initial covariate coefficients (numeric vector of length p). * **theta**: Initial spatial scale parameter (scalar initialized to 1.0). * **sigma_y_sq**: Initial residual variance (scalar initialized to 1.0).

sample_params_normal

Executes a conjugate Normal-Inverse-Gamma Gibbs sampling step for standard linear regression parameters and observation noise variance conditional on the current draw of ζ .

Usage

```
sample_params_normal(
  y,
  x = NULL,
  zeta_sam,
  params,
  prior_prec = 0.001,
  a_sigma = 0.001,
```

```
b_sigma = 0.001
)
```

Arguments

- **y**: A numeric vector of length **n** containing outcome data
- **x**: A **n x p** matrix representing known fixed-effect covariate predictors. Defaults to **NULL**
- **zeta_sam**: A numeric vector of length **n** representing the current draws of ζ .
- **params**: A named list containing current parameter values (**beta**, **theta**, **sigma_y_sq**).
- **prior_prec**: Prior precision scalar for the Gaussian prior on regression parameters. Defaults to 0.001.
- **a_sigma**: Inverse-Gamma shape hyperparameter for σ_ϵ^2 . Defaults to 0.001.
- **b_sigma**: Inverse-Gamma rate hyperparameter for σ_ϵ^2 . Defaults to 0.001.

Value

A named list containing updated parameter values (**beta**, **theta**, **sigma_y_sq**).

lik_fun_normal

Evaluates the pointwise Gaussian log-likelihood matrix of continuous outcome vector **y** across all **S** Stage 1 partial posterior draws of ζ under current parameter values.

Usage

```
lik_fun_normal(y, x = NULL, zeta_ppd, params)
```

Arguments

- **y**: A numeric vector of length **n** containing outcome data
- **x**: A **n x p** matrix representing known fixed-effect covariate predictors. Defaults to **NULL**
- **zeta_ppd**: A numeric **S x n** matrix containing Stage 1 partial posterior draws of ζ .
- **params**: A named list containing current parameter values (**beta**, **theta**, **sigma_y_sq**).

Value

A numeric matrix of dimension **S x n** containing log-likelihood values.

sample_zeta_iis

Draws a single vector ζ_{sam} using Independent Importance Sampling (IIS).

Usage

```
sample_zeta_iis(log_lik, zeta_ppd)
```

Arguments

- **log_lik**: A numeric matrix of dimension **S x n** containing log-likelihood values evaluated by **lik_fun**.
- **zeta_ppd**: A numeric **S x n** matrix containing Stage 1 partial posterior draws of ζ .

Value

A numeric vector of length **n** representing the selected draw of ζ from **zeta_ppd**.

sample_zeta_ais

Performs Adjusted Importance Sampling (AIS) without spatial copula or Vecchia re-weighting.

Usage

```
sample_zeta_ais(log_lik, zeta_ppd, R = 300)
```

Arguments

- **log_lik**: A numeric matrix of dimension $S \times n$ containing log-likelihood values evaluated by `lik_fun`.
- **zeta_ppd**: A numeric $S \times n$ matrix containing Stage 1 partial posterior draws of ζ .
- **R**: An integer defining the number of discrete candidate draws resampled in the second sampling stage of AIS. Defaults to 300.

Value

A numeric vector of length n representing the selected draw of ζ from `zeta_ppd`.

transform_to_z

Transforms candidate ζ draws from their original scale into standard Gaussian latent variables $Z \sim N(0,1)$. Uses a continuous piecewise-linear Probability Integral Transform (PIT) via empirical CDF interpolation.

Usage

```
transform_to_z(zeta_mat, sorted_zeta_ppd)
```

Arguments

- **zeta_mat**: A numeric matrix of dimension $R \times n$ containing R candidate draws across n spatial locations.
- **sorted_zeta_ppd**: A numeric matrix of dimension $S \times n$ containing sorted Stage 1 partial posterior draws for each location.

Value

A numeric matrix of dimension $R \times n$ containing transformed standard normal Z-scores.

sample_zeta_ais_copula_vecchia

Executes two-layer Adjusted Importance Sampling (AIS). Layer 1 samples candidate draws based on point-wise likelihoods; Layer 2 adjusts proposal weights using a spatial Gaussian copula density evaluated via the Vecchia approximation.

Usage

```
sample_zeta_ais_copula_vecchia(  
  log_lik,  
  zeta_ppd,  
  sorted_zeta_ppd,  
  locs,  
  covparms,
```

```
NNarray,  
R = 300,  
m = 20  
)
```

Arguments

- **log_lik**: A numeric matrix of dimension $S \times n$ containing pointwise log-likelihood values evaluated by `lik_fun`.
- **zeta_ppd**: A numeric matrix of dimension $S \times n$ containing Stage 1 partial posterior draws.
- **sorted_zeta_ppd**: A numeric matrix of dimension $S \times n$ containing column-sorted Stage 1 draws.
- **locs**: A numeric matrix of dimension $n \times 2$ specifying spatial coordinates.
- **covparms**: A numeric vector specifying spatial covariance parameters (variance, range, smoothness, nugget).
- **NNarray**: An integer matrix of dimension $n \times m$ containing pre-computed ordered nearest-neighbor indices generated by `GpGp::find_ordered_nn`.
- **R**: An integer specifying the number of draws taken from the proposal distribution for AIS. Defaults to 300.
- **m**: An integer specifying the maximum number of nearest neighbors in the Vecchia approximation. Defaults to 20.

Value

A numeric vector of length n representing the selected ζ_{sam} for the current MCMC iteration.

init_params_negbin

Generates initial values for regression parameters (β, θ) and the Negative Binomial size parameter r .

Usage

```
init_params_negbin(y, x = NULL, zeta_ppd)
```

Arguments

- **y**: A numeric vector of length n containing outcome count data
- **x**: A $n \times p$ matrix containing known fixed covariates. Defaults to `NULL`
- **zeta_ppd**: A numeric matrix of dimension $S \times n$ containing Stage 1 partial posterior draws.

Value

A named list containing:

- **beta**: Initial covariate coefficients (numeric vector of length p).
- **theta**: Initial spatial scale parameter (scalar initialized to 1.0).
- **r**: Initial dispersion/size parameter (scalar initialized to 2.0).

lik_fun_negbin

Evaluates the pointwise log-likelihood matrix of Negative Binomial outcomes across all partial posterior draws of ζ and current parameter values.

Usage

```
lik_fun_negbin(y, x = NULL, zeta_ppd, params)
```

Arguments

- **y**: A numeric vector of length **n** containing outcome count data
- **x**: A **n** × **p** matrix containing known fixed covariates. Defaults to **NULL**.
- **zeta_ppd**: A numeric matrix of dimension **S** × **n** containing Stage 1 partial posterior draws.
- **params**: A named list containing current parameter values (**beta**, **theta**, **r**).

Value

A numeric matrix of dimension **S** × **n** containing log-likelihood values.

slice_sample_r

Updates the Negative Binomial dispersion parameter **r** using a 1D univariate slice sampler under a Cauchy-like prior $p(r) \propto (1 + r^2 e_0)^{-1}$

Usage

```
slice_sample_r(r_curr, y, eta, e0 = 0.01, a_r = 0.001, b_r = 100)
```

Arguments

- **r_curr**: Numeric scalar representing the current state of **r**.
- **y**: A numeric vector of length **n** containing outcome count data
- **eta**: A numeric vector of length **n** containing current linear predictor values $\eta = X\beta + \theta\zeta$.
- **e0**: Prior hyperparameter scale factor. Defaults to 0.01.
- **a_r**: Numeric lower bound for **r**. Defaults to 0.001.
- **b_r**: Numeric upper bound for **r**. Defaults to 100.

Value

A numeric scalar representing the updated sample of **r**.

sample_params_negbin

Implements a Gibbs sampler step for Negative Binomial model parameters. Uses Polya-Gamma latent auxiliary variables ($\omega_i \sim PG$) to sample regression parameters (β, θ) conditionally from Gaussian distributions, followed by a slice sampling step for dispersion parameter **r**.

Usage

```
sample_params_negbin(  
  y,  
  x = NULL,  
  zeta_sam,  
  params,  
  prior_prec = 0.001,  
  e0 = 0.01,  
  a_r = 0.001,  
  b_r = 100,  
  r_curr = 1,  
  eta = rep(0, length(y))  
)
```

```
b_r = 100
)
```

Arguments

- **y**: A numeric vector of length **n** containing outcome count data
- **x**: A **n x p** matrix containing known fixed covariates. Defaults to **NULL**.
- **zeta_sam**: A numeric vector of length **n** representing the current sampled draws of ζ .
- **params**: A named list containing current parameter values (**beta**, **theta**, **r**).
- **prior_prec**: Prior precision scalar for Gaussian regression coefficients. Defaults to 0.001.
- **e0**: Prior hyperparameter scale factor. Defaults to 0.01.
- **a_r**: Numeric lower bound for **r**. Defaults to 0.001.
- **b_r**: Numeric upper bound for **r**. Defaults to 100.

Value

A named list containing updated parameter values (**beta**, **theta**, **r**).

setup_copula_cache

Pre-computes and caches persistent structures required for Layer 2 spatial Gaussian copula evaluations in `AIS_copula`. This includes sorting Stage 1 draws for continuous empirical CDF mapping, constructing Vecchia nearest-neighbor arrays, and fitting spatial covariance parameters with standardized Z-space variance and an explicit nugget parameter floor to prevent spatial over-smoothing.

Usage

```
setup_copula_cache(zeta_ppd, locs, m_vecchia = 20, nugget_prop = 0.10)
```

Arguments

- **zeta_ppd**: A numeric matrix of dimension **S x n** containing Stage 1 partial posterior draws.
- **locs**: A numeric matrix of dimension **n x 2** specifying spatial coordinates.
- **m_vecchia**: An integer defining the maximum number of nearest neighbors utilized in the Vecchia approximation for spatial covariance likelihood calculation. Defaults to 20.
- **nugget_prop**: A numeric scalar in $[0, 1)$ defining the proportion of total Z-space marginal variance allocated to the nugget floor. Defaults to 0.10 (10% nugget). used to prevent over-smoothing.

Value

A named list containing:

- **sorted_zeta_ppd**: A numeric matrix of dimension **S x n** containing sorted Stage 1 partial posterior draws for each location used by `transform_to_z` for fast interpolated Probability Integral Transforms (PIT).
- **NNarray**: An integer matrix of dimension **n x m** containing ordered nearest-neighbor indices generated by `GpGp::find_ordered_nn`.
- **covparms**: A numeric vector of specifying spatial covariance parameters (variance, range, nugget).